

MANISH KUMAR

Software Development Engineer II | Full-Stack | 25M+ Users | Distributed Systems

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Key Impact: **\$144K annual cost savings** • **75% faster response times** • **25M+ monthly active users** • **100% registration recovery** • **40% support ticket reduction**

TECHNICAL SKILLS

Languages	Java 17, JavaScript, C++, Go
Frameworks	Spring Boot, Spring WebFlux, React.js, Redux, Node.js, REST APIs, Microservices
Databases & Cache	DynamoDB, Redis (ElastiCache), MySQL, MongoDB, S3-backed TTL Cache
Cloud & DevOps	AWS (ECS Fargate, S3, Secrets Manager, CloudWatch), Docker, Terraform, Jenkins, Akamai CDN
Observability	Splunk, SignalFx (APM/Metrics/Traces), Grafana, Structured Logging, Distributed Tracing
Architecture	Reactive Programming, Event-Driven Design, JWT Authentication, Rate Limiting, Circuit Breaker, Cache-Aside

PROFESSIONAL EXPERIENCE

Nike, Inc. — Community Experience Platform (CXP)

Bengaluru, India

Software Development Engineer II (Full-Stack)

Jan 2025 – Present

Stack: Java 17, Spring WebFlux, Redis, DynamoDB, AWS ECS Fargate, Terraform, Jenkins, Akamai, SignalFx, Splunk, Docker

- Bot Protection & Cost Optimization:** Engineered multi-layered bot protection using Redis caching, rate limiting, and idempotency keys in Spring WebFlux, reducing WAF infrastructure by 40% (**\$144K annual savings**), slashing API traffic by 70%, and improving response times by 75% (320ms → 80ms) for Nike's event platform serving 25M+ monthly users.
- Fault-Tolerant Registration:** Designed DynamoDB-based queue with asynchronous processing to handle third-party (Eventtia) API failures, achieving **100% registration recovery** during outages by implementing conditional writes for race condition handling, duplicate prevention via conditional puts, and exponential backoff retry with status consistency checks.
- Intelligent Error Code Mapping:** Built error code determination system translating unstructured third-party API errors into 5 structured error codes with configuration-driven JSON mapping and custom exception hierarchy, **reducing support tickets by 40%** and enabling frontend to display actionable, localized user messages across US/EU regions.
- Multi-Layer Caching Architecture:** Architected three-tier caching strategy — Akamai CDN (edge), Redis ElastiCache (Pairwiseld 30-day TTL, attendee status 60s), and in-memory (translations, feature flags) — with event-driven cache invalidation on capacity errors, achieving **78% cache hit rate** and enabling 3x peak load handling without scaling.
- Conditional Registration:** Diagnosed and resolved Eventtia API inconsistency where rejected registrations were only returned via query parameters, standardizing API integration to achieve 100% consistency in displaying approved/rejected registrations and **reducing user confusion by 78%**.
- Security & Multi-Region Deployment:** Architected JWT-based shim layer for Nike-to-Eventtia authentication reducing integration overhead by 60%; rolled out multi-region deployment (US/EU) via Terraform, S3, Secrets Manager, and environment profiling with automated secret rotation.
- Observability & Reliability:** Led observability enhancements by configuring SignalFx sidecars with structured logging and ThreadContext propagation in reactive flows, increasing log accuracy by 60% and accelerating incident resolution by 25%; set up CloudWatch alarms and auto-scaling policies.

Samsung Semiconductor India Research

Bengaluru, India

Associate Staff Engineer — Senior Software Engineer

Jul 2021 – Oct 2024

Stack: Java, Spring Boot, Spring Security, React.js, JavaScript, Docker, Grafana, Kafka, MySQL, Inno Setup

- Architected in-house performance profiler and analysis tools using Java, Spring Boot, and React, eliminating external vendor dependency and achieving **\$70,000 annual cost savings**.
- Containerized UI and API servers with Docker and docker-compose, establishing CI/CD pipelines that **cut deployment time by 50%** and reduced environment-specific bugs by 70% across 15+ developer machines.
- Built authentication service using Spring Security with role-based access control for internal dashboard; implemented front-end features using React.js delivering a seamless interface across 5 departments.
- Optimized telemetry data handling, boosting sequential read/write speeds by 8% and random read/write by 5%; increased unit test coverage from 36% to 60%, reducing production bugs by 25%.
- Led sprint planning, cross-team demos, and onboarding sessions for 5+ junior developers, accelerating feature rollout by 20% and reducing new-hire ramp-up time by 30%.

PROJECTS

ShopStream — Full-Stack E-Commerce Platform

[GitHub](#)

Stack: Java 17, Spring Boot, React.js, Redux, PostgreSQL, Redis, Stripe API, Docker, AWS S3, JWT, Kafka

- Built a production-grade e-commerce platform with **microservices architecture** — Product Catalog, Cart, Order, Payment, and Notification services communicating via Kafka event-driven messaging and REST APIs.
- Implemented **Stripe payment integration** with idempotent checkout, order state machine (CREATED → PAID → SHIPPED → DELIVERED), and webhook-based payment confirmation ensuring zero duplicate charges.
- Designed **Redis-backed cart & session management** with TTL-based expiry, product catalog caching achieving sub-50ms response times, and PostgreSQL for transactional order/inventory data with optimistic locking to prevent overselling.
- Built responsive React storefront with Redux state management, JWT authentication with refresh token rotation, role-based access (Admin/Customer), and image uploads to AWS S3 with CDN delivery.

EDUCATION

Jadavpur University — Bachelor of Engineering in Information Technology

Jul 2017 – Apr 2021 | Kolkata, India

Coursework: Data Structures & Algorithms, OOP, Databases & SQL, Operating Systems, Distributed Systems, Web Development, Computer Networks